

Felipe J. Quezada-Escalona
Associate Professor, Department of Economics
Universidad de Concepción
Victoria 471, Concepción, Chile
felipequezada@udec.cl

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Dr. Sunny L. Jardine
Editor-in-Chief, *Marine Resource Economics*
University of Washington

Dear Dr. Jardine:

I am pleased to submit the manuscript *Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries* for consideration as an Article in *Marine Resource Economics*.

Climate-driven shifts in productivity reach fleets unevenly when species portfolios differ and vessel range determines which stocks each fleet can access. The paper measures this asymmetry for Chile’s Central-South small pelagic fishery (anchoveta, sardine, jack mackerel) under a fixed industrial–artisanal allocation with active cessations. I couple a Bayesian state-space stock-dynamics model fit to 2000–2024 assessments with a vessel-level trip equation. Under a six-model CMIP6 ensemble, projected effort falls by 16–18% for the artisanal fleet but only 0.8–1.0% for the industrial fleet—an asymmetry of seventeen to one, driven by artisanal exposure to the two coastal stocks whose climate shifters are large and identified. Because the artisanal fleet’s restricted range prevents access to offshore jack mackerel when coastal stocks collapse, policies that relax technological constraints—vessel upgrading, port infrastructure, weather services—may be more effective than reforms focused on quota allocation alone.

The submission consists of the main manuscript (32 double-spaced pages), an Online Appendix (six sections A–F), and this cover letter. Replication code is openly available at <https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest>. Vessel-level logbook records are accessed under restricted-access IFOP agreement and are not redistributed.

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AI disclosure. Generative AI tools were used during manuscript preparation to assist with editing for clarity, concision, and English-language polish, and to draft and refine portions of the prose. All analyses, code, data interpretation, and substantive conclusions are my responsibility, and AI outputs were verified for accuracy before inclusion.

This work has not been previously published nor is it under consideration elsewhere.

Sincerely,

Felipe J. Quezada-Escalona